JOMHACK
INSPIRE INNOVATION

PYTHON
BOOTCAMP

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STRING MANIPULATION

String Creation:

- Single Quote (one liner) rarely used
- Double Quote (one liner)
- Triple Quote (multi liner) used in a paragraph

```
single_quote = 'Hello'
double_quote = "World"
triple_quote = """Multi-line
string"""
```

STRING MANIPULATION

String indexing and slicing:

- Indexing: number(position) for each character
- Slicing: grabbing multiple characters [start:stop:step]

Index always start from 0

```python
text = "Python Programming"

print(text[0])      # (first character)
print(text[-1])     # (last character)
print(text[0:6])    # (slice 0 to 5)
print(text[:6])     # (from start to 5)
print(text[7:])     # (7 to end)
```

STRING MANIPULATION

String Methods:

- len(): Length
- strip(): remove space
- upper(): UPPERCASE
- lower(): lowercase
- title(): Titlecase
- replace(): (“old”,”new”)

```python
name = " bob the builder "

print(len(name))                    # Length
print(name.strip())                 # Remove whitespace
print(name.upper())                 # Uppercase
print(name.lower())                 # Lowercase
print(name.title())                 # Title case
print(name.replace("bob", "jane"))  # Replace
```

STRING MANIPULATION

String Formatting:

- f-string Preferably
- str.format()
- %-formatting

```python
name = "John Doe"
age = 30

message_1 = f"My name is {name} and I am {age} years old."          # f-strings
message_2 = "My name is {} and I am {} years old".format(name, age) # str.format()
message_3 = "My name is %s and I am %d years old." % (name, age)    # %-formatting

print(message_1)
print(message_2)
print(message_3)
```

STRING MANIPULATION

Exercise:

1. Build a simple text analyzer that counts words, characters, and sentences in a given text.

```
text = """Python is a powerful programming language. It's easy to learn and versatile!
You can use Python for web development, data science, and automation. The syntax is clean and readable.
This makes Python perfect for beginners and experts alike."""
```